

CV and full list of publications

Martina Seveso

Postdoctoral Researcher | Cognitive Neuroscience | Behavioural & Physiological Data Science

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OSF: <https://osf.io/bhwg8/> | Research Gate: [Martina Seveso](#)

RESEARCH PROFILE

Cognitive Neuroscientist and Postdoctoral Researcher specialising in multisensory perception, object categorisation, and neuroimaging (fMRI) with a PhD in Experimental Cognitive Psychology & Neuroscience. My research investigates how the brain integrates visual, haptic, and auditory information to form object representations and categories, combining behavioural, neuroimaging, and VR paradigms. Proficient in Python and R; experienced with Unity/C# VR development, physiological data collection (EEG, GSR/EDA, eye tracking), mixed-effects models, survival analysis, clustering, and Bayesian approaches. Strong commitment to open science and reproducible analysis pipelines.

RESEARCH INTERESTS

Multisensory perception • Sensory integration • Object categorisation & neural representation • Haptic & visual cognition • fMRI & EEG • Virtual reality as experimental tool • Ageing & perception • Computational & Bayesian approaches to cognition

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

10/2025 – Present

Trinity College Institute of Neuroscience (TCIN), Dublin, Ireland

- Designed and led fMRI experiments on visual and haptic object perception and categorisation across younger and older adults: full pipeline from task design, data collection, fMRI data acquisition to statistical analysis
- Built reproducible Python data-analysis pipelines for neuroimaging (nilearn) and behavioural data; applied LME and GLM models to large datasets
- Applied advanced statistical modelling: linear mixed effects models, survival analysis, k-means clustering in R
- Co-supervised research assistants and interns; disseminating findings at Eurohaptics 2026 (Siena) and IMRF 2026 (Genova)

Teaching Assistant - Research Methods & Statistics (Python Labs)

09/2023 – 09/2024

Trinity College Dublin, School of Psychology, Dublin, Ireland

- Delivered Python data analysis laboratory sessions to undergraduate cohorts; developed teaching materials in scientific computing and statistics

Cognitive Psychology & UX Research Intern

04/2021 – 10/2021

SR Labs: The Eye Tracking Company, Milano, Italy

- Designed UX eye-tracking experiments; built automated pipelines for pupillometry and gaze data collection in adults and children using Tobii Pro Lab eye tracking (both lab based and glasses, leading to an immersive user experience)
- Hands-on experience with physiological hardware, data collection software, and time-series analysis

Research Intern – Haptics & Virtual Prototyping Laboratory

03/2019 – 08/2019

Politecnico di Milano, Department of Mechanics, Milano, Italy

- Designed Unity-based VR experiments to simulate commercial products and human-robot interaction (HRI), including a study examining how robot movement kinematics influenced participants' physiological stress responses (SCR/EDA); collected and analysed physiological and behavioural data
- Synchronised behavioural and physiological data streams

EDUCATION

PhD, Experimental Cognitive Psychology & Neuroscience

2021 – 2025

Trinity College Institute of Neuroscience, Dublin, Ireland

Thesis: "Multisensory processing: the formation of object categories." Supervised by Prof. Fiona Newell

- Designed and implemented VR behavioural paradigms: immersive 3D object categorisation tasks with real-time haptic, visual, and auditory feedback
- Collected and synchronised multimodal data: behavioural data, eye tracking, VR positional/movement data, and reaction times
- Advanced statistical analysis: LME, survival analysis, k-means clustering in R; fMRI and EEG design and analysis; 6 peer-reviewed publications.
- Shortlisted 2nd for poster prize, IMRF 2025, Durham

Summer School: Human Factors and Virtual Reality

2021

Università degli Studi di Milano-Bicocca, Milano, Italy

- Human factors principles; Unity3D labs; immersive technology for behavioural and applied research

MSc, Applied Experimental Psychological Sciences (Summa cum laude)

2017 – 2020

Università degli Studi di Milano-Bicocca, Milano, Italy

Thesis: "The Multisensory Experience of Sweet Food: An Empirical Study" (FoodNet Project). Supervised by Prof. Alberto Gallace

- Specialisation in Cognitive Psychology & Neuroscience; statistical analysis, Python and R programming, experimental design

BSc, Psychological Sciences and Techniques

2013 – 2017

Università degli Studi di Milano-Bicocca, Milano, Italy

Thesis: Psychophysiological Correlates of Visual-Tactile Consumer Product Exploration. Supervised by Prof. Alberto Gallace

- EEG & GSR data collection and analysis

TECHNICAL SKILLS

Programming	Python (advanced: numpy, pandas, scipy, sklearn, matplotlib, nilearn), R (advanced: lme4, ggplot2, survival, tidyverse), MATLAB, C# (basic for Unity), JavaScript (basic), GitHub
VR & Experimental Systems	Unity (VR experiment design and integration), PsychoPy, Pavlovia; real-time data collection
Physiological & Sensing	EEG and GSR/EDA (data collection and data cleaning in BSc thesis), Pupillometry (data collection and analysis in SR Labs, Tobii Pro Lab), fMRI (data collection, preprocessing and analysis in Python); experienced synchronising multimodal data streams
Statistical & Computational	LME models, Bayesian causal inference, survival analysis, k-means clustering, GLM, dimensionality reduction; reproducible pipelines for open science
3D & Movement	3D stimulus design and printing in Blender, Fusion 360, AutoCAD; VR kinematic data handling and implemented webcam-based hand tracking via mediapipe (python)
Open Science	OSF, GitHub, preregistration, GDPR-compliant data management

CERTIFICATIONS

- Creating Perceptual and Behavioral Experiments with Unity and VR, York University
- Bayesian Causal Inference, in person workshop at Durham University
- Machine Learning, Stanford University (Andrew Ng)
- Computational Neuroscience: Neuronal Dynamics of Cognition, EPFLx
- Computational Neuroscience, University of Washington
- Complete Neural Signal Processing and Analysis, Mike X Cohen, Radboud University
- Neuroscience and Psychology: Electroencephalography (EEG), University of Leeds
- SPM Course for fMRI/VBM, in person workshop at University College London
- Principles of fMRI (1 & 2) / Neuroscience and Neuroimaging Specialization, Johns Hopkins University
- Tobii Pro Lab for Research, Tobii
- Data Visualisation & Dashboarding with R Specialisation, Johns Hopkins University
- Python for Data Science, AI & Development, IBM
- Research Integrity Training, Epigeum | Data Protection / GDPR, Trinity College Dublin

PUBLICATIONS

Peer-Reviewed Articles

Seveso, M.A., O'Dowd, A., Hirst, R.J., & Newell, F.N. Motion, but not sound, affects the inter-object perceptual similarity of 3D novel objects. *Journal of Vision* (*in prep*)

Seveso, M.A., Kidney, E., & Newell, F.N. (*under review*, 2026). Evidence for a Haptic Over Vision Bias in the Categorisation of Novel Objects using Virtual Reality. *Transactions on Applied Perception*

Seveso, M.A., Hirst, R.J., O'Dowd, A., Camponogara, I., & Newell, F.N. (2026). Visual and tactile motion cues enhance the categorisation of novel object shapes. *Experimental Brain Research*, 244(4), 48

Hirst, R., Roberts, K., **Seveso, M.A.**, O'Dowd, A., Peirce, J., & Newell, F.N. (2026). Delivering tactile stimuli via mobile browsers. *Behavior Research Methods*, 58(4), 106

O'Dowd, A., Hirst, R.J., **Seveso, M.A.**, McKenna, E., Newell, F.N. (2024). Generalisation to novel exemplars of learned shape categories. *Psychonomic Bulletin & Review*

Newell, F.N., McKenna, E., **Seveso, M.A.**, et al. (2023). Multisensory perception constrains the formation of object categories. *Philosophical Transactions of the Royal Society B*

Full list: Research Gate - [Martina Seveso](#), ORCID - <https://orcid.org/0000-0001-6566-4578>

SELECTED CONFERENCE PRESENTATIONS

- **Seveso M.A.** et al. Haptic Category Learning and Cross-Modal Transfer. Eurohaptics 2026, Siena [poster]
- **Seveso M.A.** et al. Flexible Multisensory Category Representations. IMRF 2026, Genova [poster]
- **Seveso M.A.** et al. Haptic Cues enhance online categorization. ECVF 2025, Mainz [poster]
- **Seveso M.A.** et al. Haptic Cues enhance online categorization. IMRF 2025, Durham [poster] * **Shortlisted 2nd for poster prize**
- **Seveso M.A.** et al. Object movement or sound and inter-item perceptual similarity. Psychonomic Society 2024, New York [poster]
- **Seveso M.A.** et al. Motion affects inter-object perceptual similarity. OPAM 2024, New York [oral presentation]
- **Seveso M.A.** et al. Motion affects inter-object similarity. EPS Nottingham 2024 [poster]
- **Seveso M.A.** et al. Cross modal interactions in categorical perception. IMRF 2023, Brussels [poster]

LANGUAGES

Italian (mother tongue) | English (fluent) | French (beginner, actively improving for francophone Switzerland)

PUBLIC ENGAGEMENT & SERVICE

- European Researcher Night (2022–2024), Dublin, Ireland
- Touch in perspective: Have we lost sight of our sense of touch? SHUF (2021), Dublin
- European Researcher Night (2019), Milan, Italy
- Volunteer, Comunità Nuova Onlus – food distribution during COVID-19 (2021), Milan

REFEREES

- Prof. Fiona N. Newell, Trinity College Dublin | FNEWELL@tcd.ie
- Dr. Alan O'Dowd, Trinity College Dublin | odowda1@tcd.ie
- Dr. Rebecca J. Hirst, Trinity College Dublin | HIRSTR@tcd.ie